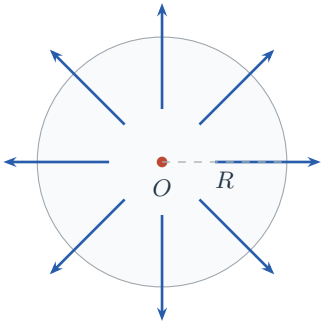
M1-53 球面通量核对散度的体积分



$\mathbf{A}=\mathbf{r}$；球面外法向为 $\mathbf{e}_r$

$$\nabla\cdot\mathbf{r} = 1+1+1 = 3$$

$$\Phi = 4\pi R^2\cdot R = 4\pi R^3$$

$$\int_V 3\,dV = 3\frac{4\pi R^3}{3} = 4\pi R^3$$

任意光滑闭区域也满足 $\Phi = 3V$。